



Lecture Notes on Electricity, Magnetism and Modern Physics

Pre-requisite: Credit pass in O/level Physics

DEPARTMENT OF PHYSICS
FEDERAL UNIVERSITY LAFIA



Course Outline

Electricity: The electric circuit, arrangement of resistors in a circuit, Ohm's law, internal resistance of a cell, the e.m.f., Kirchhoff's laws, potentiometer, the wheat-stone bridge, electrostatics, electric charges, methods of charging bodies, Coulomb's laws, electrolysis, example of electrolysis, application of electrolysis, Faraday's laws of electrolysis, alternating current (A.C.), r.m.s., phasor representation of A.C., resistance, capacitance, inductance, series containing R, L and C, resonance in R, L and C circuit.

Magnetism: Magnetic field and forces on a conductor, Hall Effect, electromagnetic induction, dynamo and generator, transformer and magnetic properties of matter.

Modern Physics: Models of the atom, radioactivity: fusion, fission and half-life, photoelectric effect and uncertainty principle.

Some Useful Textbooks

1. University Physics with Modern Physics by Young, H. D., Freedman, R. A. and Ford, A. L.
2. New School Physics by Anyakoha, M. W.
3. College Physics by Bueche, F. J. and Hetch, E.
4. Electricity and Magnetism by Crowell, B.

CHAPTER ONE

1.1 Introduction

PHY121 (electricity, magnetism and modern physics) is a three (3) credit fundamental level course to be taken at second semester of each academic sessions.

The course comprises 3 sections, which involve elementary principles of Electricity, Magnetism and Modern Physics. This course material have been developed in such a way that a student with at least a credit pass in O/level Physics will follow quite easily. The compulsory pre-requisites for the course is therefore a credit pass in O/level Physics. It is also strongly advised to have adequate grasp of Further Mathematics or Applied Mathematics.

There are regular tutorial classes that are linked to the course. You are strongly advised to attend these sessions regularly. Details of time will be given to you during the course of lecture.

1.2 Course Aims

The aim of this course is to introduce the basic elementary principle and application of Electrical Energy and its relationship with Magnetism. During the course of the lecture, you will learn that an electric field is always associated with a magnetic field and vice versa.

Towards the end of the course you will be introduced into some aspects of Modern Physics.

1.2 Assessment

There are three aspect of the assessment of the course. First is made-up of series of assignments, second consists of written test at the end of the course work and third is the written examination. In tackling the assignments, you are expected to apply information,

knowledge and techniques you gathered during the course. The assignments must be submitted at Physics Lecturers office (LPH2) for formal assessment in accordance with the deadlines stated in the assignment file.

The assignments you submitted for assessment will count for 15% of your total course work. At the end of the course work you will sit for a pre-exam (Test). This test will count for 15% of your total course work. Also, you will sit for final examination of about three hours duration. This examination will count for 70 percent of your total course mark.

Assignment	Marks
Assignments 1- 4	4 assignments, best 3 marks of the 4 count at 5 % each – 15 % of the course marks (C.A. 1)
Pre-exam (Test)	15 % of overall course marks (C. A. 2)
End of semester examination	70 % of overall course marks

Should you have any problem [as regard the course](#), do not hesitate to come to LPH2 or contact me by email add: fbjames11@physicist.net.

I wish you a splendid study time.

Solutions to all worked examples shall be done in class during the course of lectures. It is therefore strongly recommended that you attend lectures.

CHAPTER TWO

2.1 Electricity

Electricity is the set of physical phenomena associated with the presence and flow of electric charge. Electricity gives a wide variety of well-known effects, such as lightning, static electricity, electromagnetic induction and electrical current. In addition, electricity permits the creation and reception of electromagnetic radiation such as radio waves.

Electricity can be generated from

- Chemical energy.
- Heat energy (The thermoelectric effect).
- Mechanical energy.
- Solar energy.

2.2 The Electric Circuit

An electric circuit is the complete path provided for the flow of electric current. The circuit usually consists of the source of electric energy, the battery or cell, a switch or key to start or stop the current flow, a voltmeter to measure the potential difference, an ammeter to measure the current flow, a resistor or load and a rheostat to adjust the current flow. These components are connected as shown in figure below

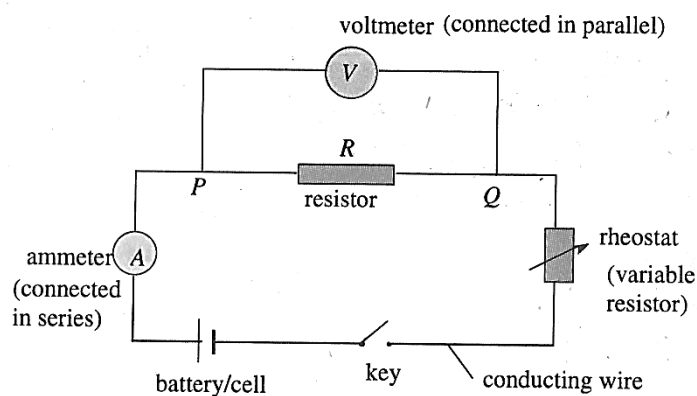


Figure 2.1

It should be noted that ammeter must always be connected in series in such a way that the current it measures flows directly through it. In order not to alter the current it

measures, the ammeter is constructed to have a low resistance. The voltmeter is constructed to have a large electrical resistance compared to the resistance across the points they are connected. They therefore take negligible amount of current.

Another important component is the resistor: A resistor is a component which is specially designed to provide a known amount of resistance in a circuit.

Homework 1: Distinguish between standard resistors and variable resistors

2.3 Arrangement of Resistors in a Circuit

There may be more than one resistor in an electric circuit. Such resistors may be connected in parallel or in series.

When the resistors are connected in end to end as shown in the figure below, they are said to be in series.

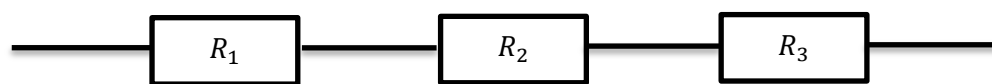


Figure 2.2

Thus, the equivalent resistance of the combination is given by

$$R_T = R_1 + R_2 + R_3$$

However, when resistors are arranged side by side such that their corresponding ends join together at two common junctions P and Q as shown in fig 2.3 below, the arrangement is said to be parallel connection of the resistors. In such arrangement, the p.d. across the resistors is the same but the current through each resistor is different.

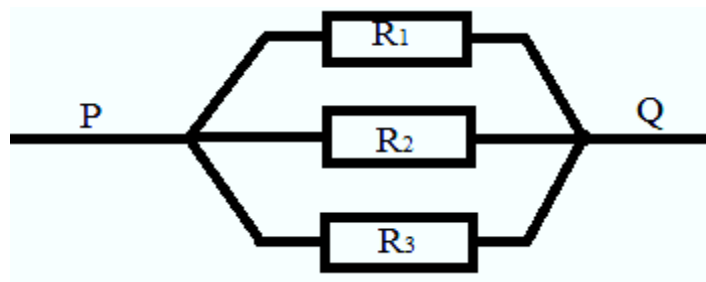


Figure 2.3

Thus, the combined or equivalent resistance is given by

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Worked example 2.1

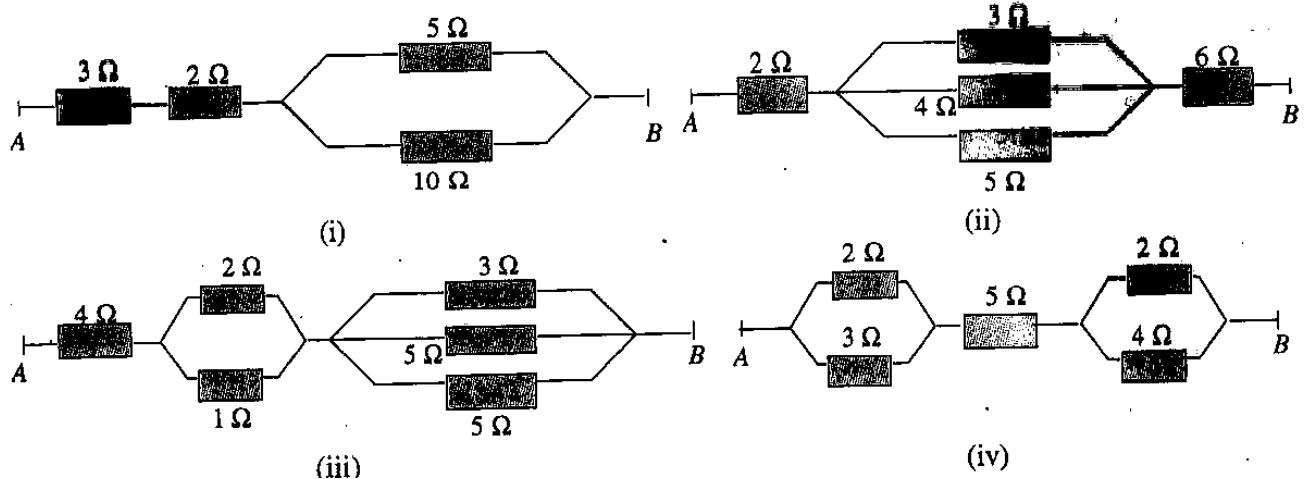
Three 3.0 ohms resistors are connected in parallel. What is the equivalent resistance?

Worked example 2.2

Four resistors of resistance 1Ω, 2Ω, 3Ω, 5Ω, are connected in series and afterwards in parallel. Calculate the combined resistance in each case.

Worked example 2.3

Calculate the combined resistance of the following combinations shown below



2.4 Ohm's Law

The current flowing through a metallic conductor (e. g. wire) is directly proportional to the potential difference across its ends, provided that temperature and other physical conditions of the conductor are kept constant. Mathematically

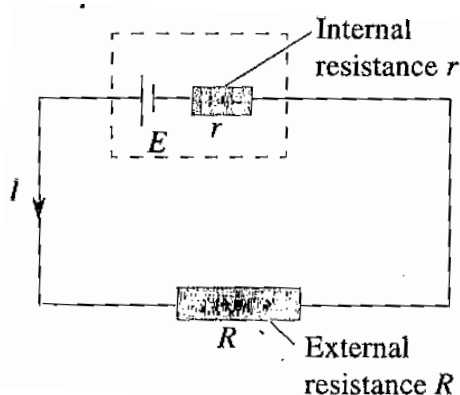
$$I \propto V$$

where I is the current and V is the potential difference (p.d.). Removing the sign of the proportionality, we have

$$\frac{V}{I} = \text{constant} \quad \text{or} \quad \frac{V}{I} = R,$$

where R is the resistance.

2.5 Internal Resistance of a Cell



The internal resistance, r , of a cell is the opposition to current flow offered by chemicals between the poles of the cell

Figure 2.4: Internal resistance of a cell

2.6 The Electromotive Force (e.m.f.)

The e.m.f. of a cell is the p.d. across its terminals when it's in an open circuit, i.e. not supplying current to an external circuit.

By Ohm's law such current is given by

$$I_{max} = \frac{E}{r}$$

According to figure 2.4 above, the cell of e.m.f., E , supplies a current $I(A)$ to an external resistance R , then the p.d. across R , known as Terminal p.d. is given by

$$E = V + v$$

where v is the p.d. across the internal resistance, r . Using Ohms law, we can re-write above equation as

$$E = IR + Ir = I(R + r) \quad \text{or} \quad I = \frac{E}{R + r}$$

It is worth mentioning that the terminal p.d., V , across the external resistance R , is less than the e.m.f., E of the cell, by the *lost voltage* (v)

$$V = E - v$$

Worked Example 2.4

A battery made up of 2 cells joined in series supplies current to an external resistance of 5 ohm. If the e.m.f. and internal resistance of each cell is 0.6V and 3ohm respectively, calculate the current flowing in the external resistor, the terminal p.d., and the lost voltage.

Worked Example 2.5

A cell can supply currents of 0.80A and 0.40A through a 2 ohm and a 5 ohm resistor respectively. Calculate the internal resistance of the cell.

Worked Example 2.6

A battery of three cells in series, each of e.m.f. 2V and internal resistance 0.5 ohm is connected to a 2 ohm resistor in series with a parallel combination of two 3 ohm resistors. Draw the circuit diagram and calculate:

- the effective external resistance
- the current in the circuit
- the lost volts in the battery
- the current in one of the 3 ohm resistors

Worked Example 2.7

A cell of e.m.f. 1.08v is in series with a resistor of resistance 80 ohm. A high resistance voltmeter put across the cell registers only 0.8V. What is the internal resistance of the cell?

2.7 Kirchhoff's Laws

Many practical resistor networks cannot be reduced to simple series-parallel combinations. Figure below (a) shows a dc power supply with e.m.f. ($\mathcal{E}_1$) charging a battery with a smaller e.m.f. ($\mathcal{E}_2$) and feeding current to a light bulb with resistance R . Also, a “bridge” circuit, used in many different types of measurement and control systems is shown in the b part.



Figure 2.5

To compute the currents in these networks, we'll use the techniques developed by the German physicist Gustav Robert Kirchhoff (1824–1887).

First, here are two terms that we will use often. A **junction** in a circuit is a point where three or more conductors meet. A **loop** is any closed conducting path. In Fig. a points a and b are junctions, but points c and d are not; in Fig. b the points a , b , c , and d are junctions, but points e and f are not. Some possible loops in these circuits have been indicated.

Kirchhoff's rules are the following two statements:

Kirchhoff's junction rule: *The algebraic sum of the currents into any junction is zero.* That is,

$$\sum I = 0 \quad (\text{junction rule, valid at any junction})$$

Kirchhoff's loop rule: *The algebraic sum of the potential differences in any loop, including those associated with e.m.f.s and those of resistive elements, must equal zero.* That is,

$$\sum V = 0 \quad (\text{loop rule, valid for any closed loop})$$

Sign Conventions for the Loop Rule

In applying the loop rule, we need some sign conventions. We first assume a direction for the current in each branch of the circuit and mark it on a diagram of the circuit.

Then, starting at any point in the circuit, we imagine traveling around a loop, adding e.m.f.s and IR terms as we come to them. When we travel through a source in the direction from $-$ to the $+$ the emf is considered to be *positive*; when we travel from $+$ to the $+$ the emf is considered to be *negative*. When we travel through a resistor in the *same* direction as the assumed current, the IR term is *negative* because the current goes in the direction of decreasing potential. When we travel through a resistor in the direction *opposite* to the assumed current, the IR term is *positive* because this represents a rise of potential. See below figure

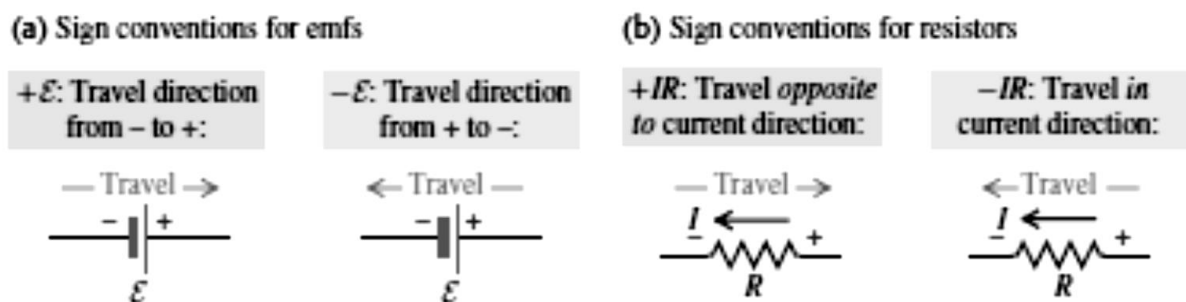


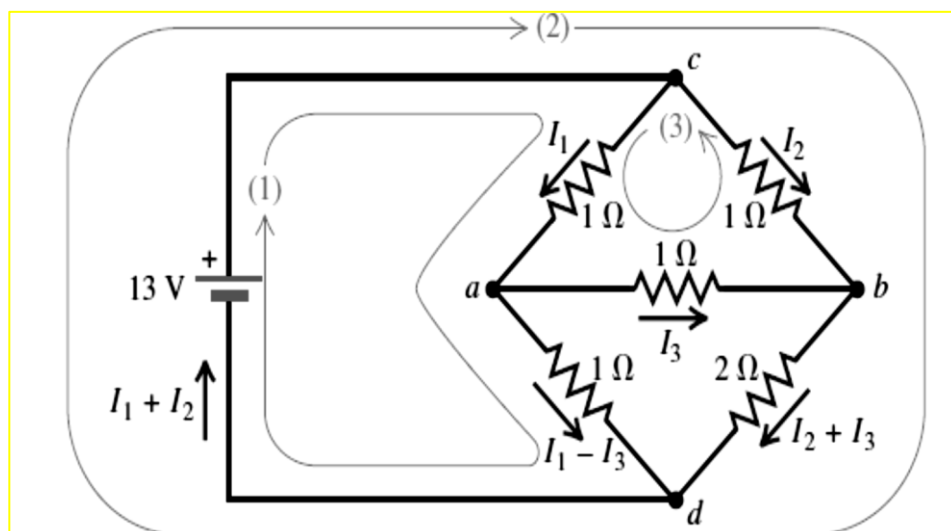
Figure 2.6

Kirchhoff's two rules are all we need to solve a wide variety of network problems. Usually, some of the e.m.f.s, currents, and resistances are known, and others are unknown. We must always obtain from Kirchhoff's rules a number of independent equations equal to the number of unknowns so that we can solve the equations simultaneously. Often the hardest part of the solution is not understanding the basic principles but keeping track of algebraic signs!

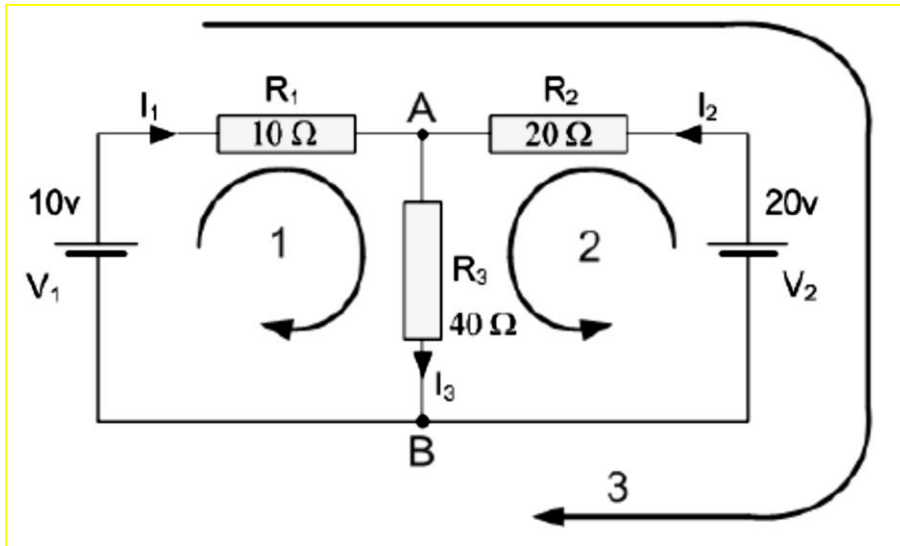
Worked Example 2.8

Figure shows a “bridge” circuit of the type described at the beginning of this section. Find the current in each resistor and the equivalent resistance of the network of five resistors.

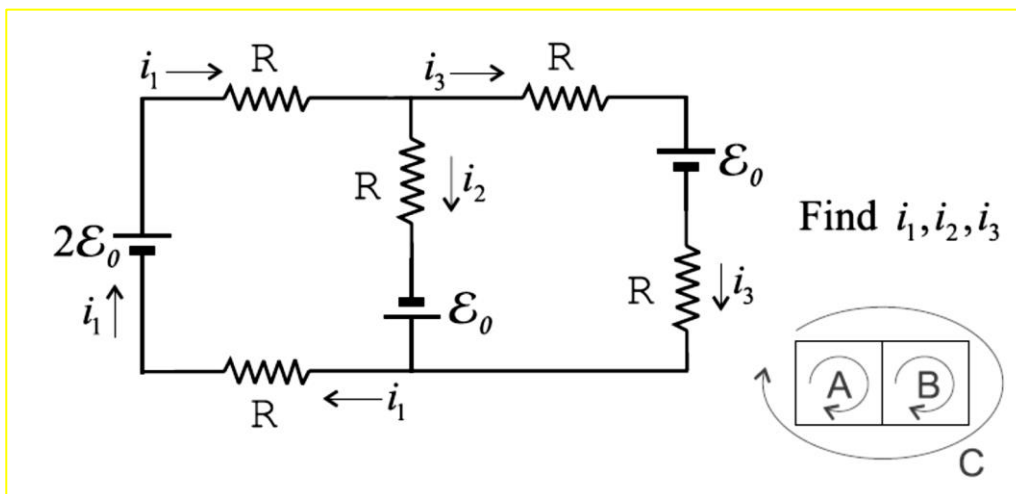
Worked Example 2.9



Determine the values of the current flowing through each of the resistors.



Worked Example 2.10



2.8 Potentiometer

The potentiometer is a device to precisely measure potential differences. It's an apparatus commonly used to measure or to compare the e.m.f.s of cells. It can also be used to compare or measure resistances. It is also used for adjusting the volume level on a radio or changing the channels of some types of television sets.

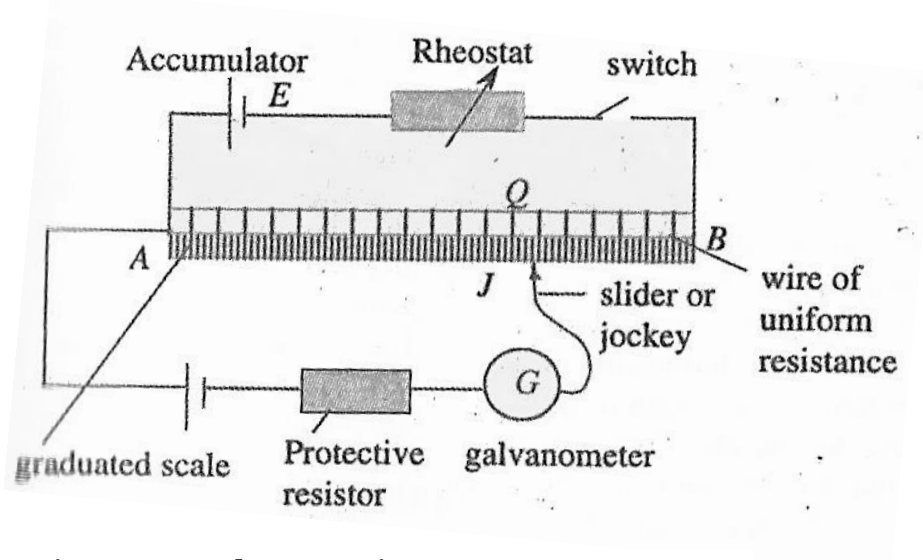


Figure 2.7: The potentiometer

To compare or measure e.m.f.s, the following relation

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

is found useful.

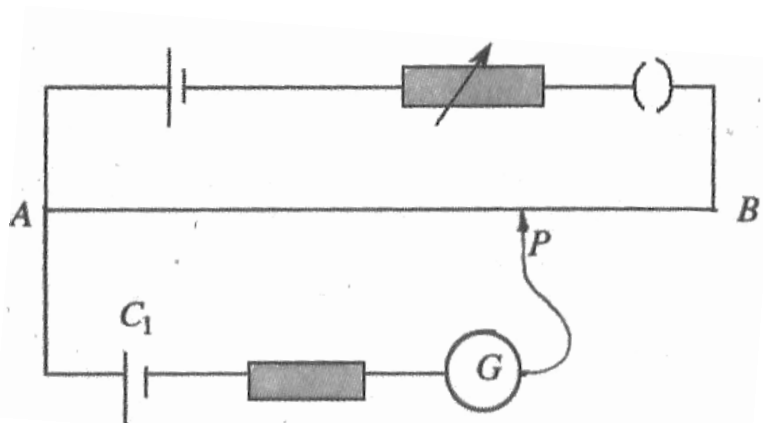


Figure 2.8: Comparison of e.m.f. of cells

E_1 denotes the e.m.f. of the cell C_1 and l_1 is the corresponding length AP. By replacing cell C_1 with another cell C_2 whose e.m.f. is E_2 , thus the corresponding new length is l_2 .

Worked Example 2.11

The current in a potentiometer is adjusted with a rheostat so that the galvanometer reads zero when a standard cell of e.m.f. 1.02volts connected to the slider intercepts 40.0cm of the potentiometer wire when a cell, C_1 of unknown e.m.f. is substituted for the standard cell, the balance point is at 49.0cm. calculate the e.m.f. of the cell

2.9 The Wheatstone bridge

One of the most accurate methods of measuring resistance is the Wheatstone bridge. The Wheatstone bridge circuit consists of four resistances arranged as shown in figure below.

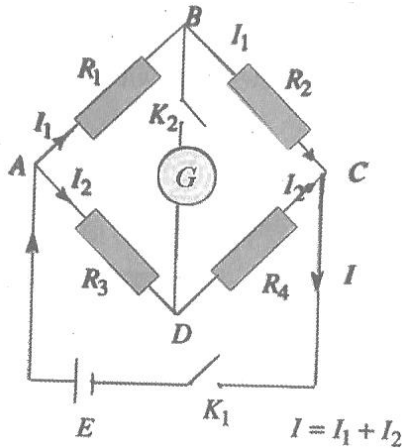


Figure 2.9: Wheatstone bridge

G is a sensitive galvanometer connected across the junctions B and D of the circuit. E is a dry cell of e.m.f. 1.5 volts which drives a current through the various arms of the bridge. Keys, k_1 and k_2 are connected to the arms of the cell and galvanometer.

Since B and D are at the same potential,

p.d. across AB = p.d. across AD,

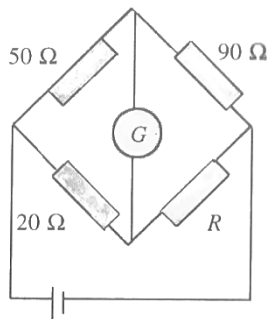
p.d. across BC = p.d. across CD,

therefore $I_1 R_1 = I_2 R_3$ and $I_1 R_2 = I_2 R_4$. Thus, the following relation can be deduced

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Worked Example 2.12

Calculate the value of R when G shows no deflection in the circuit illustrated below



CHAPTER THREE

3.1 Electrostatic

We begin our study in this chapter by examining the nature of electric charge. We'll find that charge is quantized and obeys a conservation principle. When charges are at rest in our frame of reference, they exert *electrostatic*¹ forces on each other. These forces are of tremendous importance in chemistry and biology and have many technological applications. Electrostatic forces are governed by a simple relationship known as *Coulomb's law* and are most conveniently described by using the concept of *electric field*.

There are practices in everyday life that clearly show this effect.

Examples;

- (i) Running a comb through one's hair on a dry day it will be observed that the comb is capable of attracting bits of paper and sometimes the attractive force is strong enough to suspend the paper.
- (ii) If an inflated balloon is rubbed with wool, it can stick to a wall sometimes for some hours.

Materials which behave as described above are said to be electrified or electrically charged.

3.2 Electric charges

Experiments and many others like them have shown that there are exactly two kinds of electric charge: the kind on the plastic rod rubbed with fur and the kind on the glass rod rubbed with silk. Benjamin Franklin (1706–1790) suggested calling these two kinds of charge *negative* and *positive*, respectively, and these names are still used. The plastic rod and the silk have negative charge; the glass rod and the fur have positive charge.

Two positive charges or two negative charges repel each other. A positive charge and a negative charge attract each other.

¹ Electrostatics means static electricity.

3.2 Properties of electric charges

- (a) Electric charges can neither be created nor destroyed thus they are always conserved. They can only be transferred from one object to another. Thus one object gains some amount of negative charge while the other gains an equal amount of positive charge.
- (b) Electric charge is quantized.
- (c) Like charges repel while unlike charges attract one another. This is known as the fundamental law of electrostatics.
- (d) The space surrounding a charge is referred to as its electric field. Electric lines of force leave from a positive charge and enter into a negative charge.

3.2 Method of charging bodies

- (i) **By Friction:** This involves rubbing one object on another. E.g when a glass rod is rubbed with silk, the glass rod becomes positively charged. On the other hand a plastic rod rubbed with fur becomes negatively charged. Ebonite rod rubbed with fur becomes negatively charged while Cellulose acetate strip rubbed with silk becomes positively charged while Polythelene rod rubbed with fur becomes negatively charged.
- (ii) **By contact:** This involves bringing an uncharged body in contact with a charged body. Usually the uncharged body acquires a charge opposite to that of the charging body.
- (iii) **By Electrostatic induction:** This is the process of charging a neutral body by placing a charged body near it without any contact with it.

3.2 Coulomb's Law

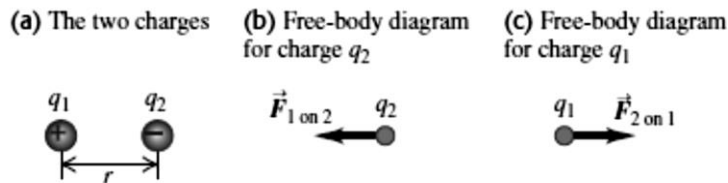
The force between two charges Q_1 and Q_2 , separated by a distance r apart is proportional to the product of the charges and inversely proportional to the square of the distance between them;

$$\text{Mathematically } F \propto \frac{|Q_1 Q_2|}{r^2} \text{ thus } F = k \frac{|Q_1 Q_2|}{r^2}$$

where k is a proportionality constant whose numerical value is taken as $\frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ N.m}^2/\text{C}^2 \cong 9.0 \times 10^9 \text{ N.m}^2/\text{C}^2$. The absolute value bars are used in the above equation because the charges q_1 and q_2 can be either positive or negative, while the force magnitude is always positive.

Worked Example 3.1

Two point charges $q_1 = +25nC$ and $q_2 = -75nC$ are separated by a distance $r = 3.0cm$ (See the figure below). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .



Worked Example 3.2

An α particle (the nucleus of a helium atom) has mass $m = 6.64 \times 10^{-27}kg$ and charge $q = +2e = 3.2 \times 10^{-19}C$. Compare the magnitude of the electric repulsion between two (“alpha”) particles with that of the gravitational attraction between them.

Worked Example 3.3

Two point charges are located on the x -axis of a coordinate system: $q_1 = 1.0nC$ is at $x = +2cm$ and $q_2 = -3.0nC$ is at $x = +4cm$. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5.0nC$ at $x = 0$?

Worked Example 3.4

Two equal positive charges $q_1 = q_2 = 2\mu C$ are located at $x = 0, y = 0.3m$ and $x = 0, y = -0.3m$ respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 40\mu C$ at $x = 0.4$ and $y = 0$.

CHAPTER FOUR

4.1 Electrolysis

Electrolysis is the chemical change in a liquid due to the flow of electric current through the liquid. Michael Faraday studied this process extensively and laid the foundation for the theory of electrolysis. Some liquid are good conductors while others are poor conductors of electricity. Good conductors are known as *electrolytes*², poor conductors are known as *non-electrolyte*.

A voltmeter is a device for studying the flow of current through a liquid.

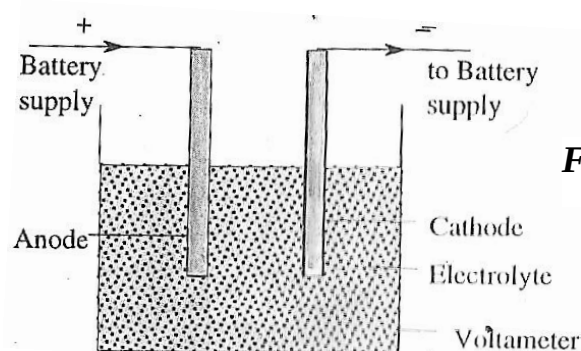


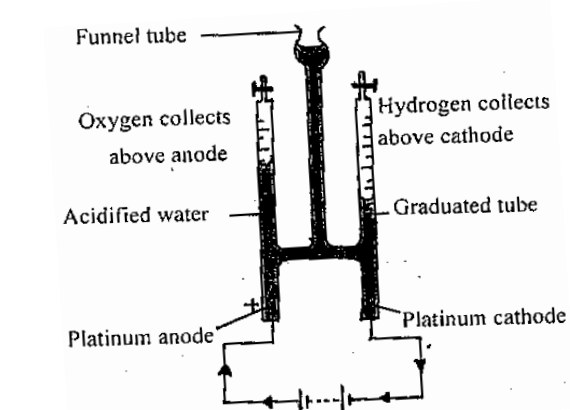
Figure 4.1: Voltmeter

It consist of the electrolyte and two electrodes namely:

- **Cathode:** electrodes at which current leaves the electrolyte.
- **Anode:** electrodes at which current enters the electrolyte.

4.2 Examples of electrolysis

4.2.1. Electrolysis of acidulated water



The experimental setup is shown in the figure below

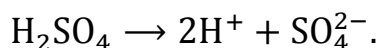
Figure 4.2: Electrolysis of water via the Hoffman

² Electrolyte is the liquid or molten substance which conducts a current and is decomposed by it.

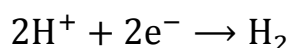
In this process, water dissociate into hydrogen ions (H^+) and hydroxyl ions (OH^-)



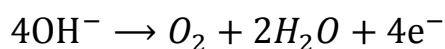
The sulphuric acid dissociates into hydrogen ions and sulphate ions



When current is passed through the platinum electrodes in acidulated water, H^+ ions drift to the cathode and are discharged there:



OH^- ions drift to the anode and are discharged more easily than SO_4^{2-} ions, resulting in the formation of oxygen and water



4.2.2. Electrolysis of copper sulphate solution with copper electrodes

The setup apparatus is shown in the figure below

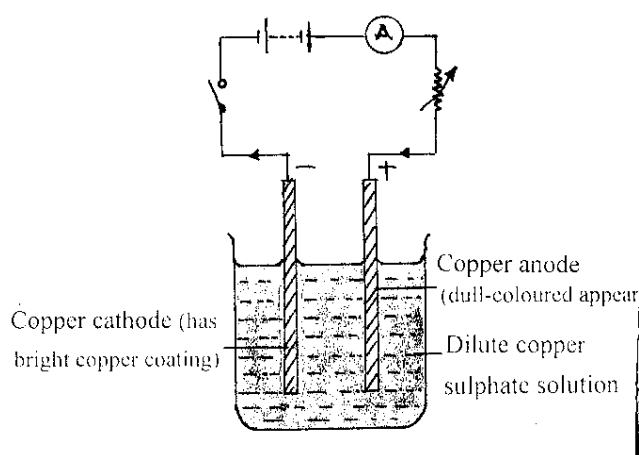
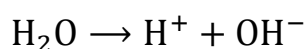
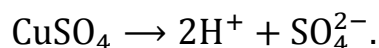
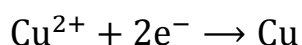


Figure 4.3: Electrolysis of copper sulphate solution

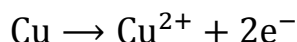
The copper sulphate in solution dissociates into copper ions and sulphate ions. The water breaks down into hydrogen ions and hydroxyl ions.



When electricity is passed, both the copper and hydrogen ions drift to the cathode where the copper is discharged in preference to the hydrogen. Hence copper is deposited at the cathode.



The SO_4^{2-} and OH^- drift to the anode but here the discharge of copper ions into solution takes precedence over the discharge of SO_4^{2-} and OH^- . Hence we have



The overall result of the process described is the flow of copper from the anode to the cathode and a flow of electrons round the external circuit. There is no net gain or loss of copper from the solution. Hence the density of the copper sulphate from the solution is unchanged throughout the electrolysis.

4.3 Applications of electrolysis

Electrolysis finds several applications in industry, for example, in the electroplating of metals, the purification of impure metals and electrolytic production of metals from their compounds.

4.4 Faraday's Laws of electrolysis

From the result of his experiments, Faraday discovered two laws of electrolysis.

Faraday's 1st law states that mass, M , of a substance liberated in electrolysis is directly proportional to the quantity of electricity Q , which has passed through the electrolyte.

Mathematically,

$$M \propto Q,$$

where Q is the quantity of electricity in coulombs but since $Q = It$, we can then say that

$$M \propto It,$$

where I is the current (amperes) and t is the time (in seconds) during which the current flows. Hence

$$M = ZIt,$$

where Z is a constant of the proportionality known as the electrochemical equivalent³ (e.c.e.) of the substance

³ The electrochemical equivalent (e.c.e.) of a substance is the mass of that substance deposited by one Coulomb of electricity in a process of electrolysis. (Unit = g/Coulomb)

Faraday's 2nd law states that masses of different substance deposited or liberated by the same quantity of electricity are directly proportional to the chemical equivalents of the substances.

Worked Example 4.1

Find the mass of a copper deposited on the cathode of a copper voltmeter if a current of 0.53A is passed through it for 30 minutes (e.c.e. of copper = $3.3 \times 10^{-4} \text{g/C}$).

Worked Example 4.2

Calculate the time in minutes required to electroplate an article of area 300cm^2 with a layer of copper 0.06cm thick if a constant current of 2A is maintained. Assumed that the density of copper is 8.8gcm^{-3} and that one coulomb liberates 0.00033 g copper.

Worked Example 4.3

A copper and a silver voltmeter are connected in series, and at the end of a period of time, 5.0g of copper was deposited, calculate the mass of silver deposited at the same time. Chemical equivalent of copper = 31.5. Chemical equivalent of silver = 1.08.

CHAPTER FIVE

A. C. circuits are circuits through which an alternating current flows. Such circuits are used extensively in power transmission, radio and television, computer technology, telecommunication and in medicine. Alternating currents are produced by time dependent alternating voltages given by the relation

$$E = E_0 \sin \omega t$$

5.1 Alternating Current

An Alternating Current (A.C.) is the one that varies sinusoidally or periodically in such a way as to reverse its direction periodically. The commonest form such an a.c. can be represented

$$I = I_0 \sin 2\pi f t = I_0 \sin \omega t$$

where I is the instantaneous current at a time t , I_0 is the maximum (or peak) value of current or its amplitude; f is the frequency and ω is the angular velocity, (ωt) is the phase angle of the current. Alternating voltage is also represented by

$$V = V_0 \sin 2\pi f t = V_0 \sin \omega t$$

V , V_0 are the instantaneous and peak (or maximum) values of the voltage or its amplitude.

5.2 Root-mean-square (rms)

Root-mean-square is that steady current which will develop the same quantity of heat in the same time in the same resistance.

Mathematically we have

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

5.3 Phasor representation of alternating current

Suppose that the voltage V across a given circuit element at the instant t is a sinusoidal function of time with angular frequency $\omega = 2\pi f$, given by

$$V = V_0 \cos (\omega t + \theta_{0v})$$

You recall that the positive constant V_0 is the amplitude of this function. This is the largest value attained by the instantaneous voltage. The constant θ_{0v} is the initial phase angle of the voltage function.

The current I in the circuit element is a sinusoidal function of the same angular frequency ω as that of the voltage V across the circuit element. The current at the instant t is therefore a function of the form.

$$I = I_0 \cos (\omega t + \theta_{0I})$$

This function has an amplitude I_0 and an initial phase angle θ_{0I} .

The angle θ between the voltage phasor and the current phasor is called the phasor difference between the voltage and the current. This phasor difference θ is the difference of the phase angles of the functions V and I :

$$\theta = (\omega t + \theta_{0V}) - (\omega t + \theta_{0I})$$

If $\theta = 0$, the voltage and the current are said to be in phase, otherwise they are out of phase.

Worked Example 5.1

The instantaneous voltage V and current I are given in SI units by

$$V = -155 \cos (377t - \pi/2)$$

$$I = 2.0 \sin 377t$$

Find the phase difference between the voltage and the current.

5.4 Resistance, Capacitance and Inductance in A.C. Circuit

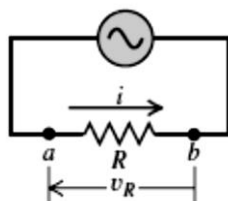


Figure 5.1: A.C. circuit with a resistor

The above figure shows a simple a.c. circuit with a resistor. The voltage and ammeter are connected to the circuit to read r.m.s. value of the voltage and current. Hence, we can write

$$I_{r.m.s} = \frac{V_{r.m.s}}{R}$$

The voltage and the current are said to be in phase or in step with each other. *This means that both attain their maximum, zero and minimum at the same instant.*

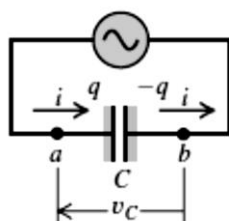


Figure 5.2: A.C. circuit with a capacitor

The above figure shows a simple a.c. circuit with a capacitor. The voltage and the current are out of phase. Thus the current is said to lead on the voltage and the voltage is said to lag on the current.

The capacitor opposes flow of current. This opposition to the flow of a.c. current offered by capacitor is known as *capacitive reactance* X_c given by the relation

$$X_c = \frac{1}{2\pi fC}.$$

When an A.C. voltage of frequency f is applied to the capacitance, C , then the following relation holds

$$V_0 = I_0 X_c$$

In other word, an Ohm's law is applies to a capacitor. In it R is replaced by X_c . Hence the unit of X_c is ohm.

Worked example 5.2

A $2\mu F$ capacitor is connected directly across a $150 V_{rms}$, 60Hz a.c. source. Find (a) the r.m.s. value of the current (b) the value of current.

The below figure shows a simple a.c. circuit with an inductor.

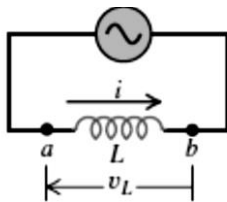


Figure 5.3: A.C. circuit with a inductor

The induced e.m.f. in the inductor L opposes the change in the current. As a result the current is delayed behind the voltage in the circuit. Then, we say that the current I lag behind V by $\frac{\pi}{2}$ radian or 90° . In other word, I and V have a phase difference of $\frac{\pi}{2}$ or 90° .

Like R and C , an inductor L opposes flow the flow of current; i.e., it has an impedance effect known as the inductive reactance X_L

$$X_L = 2\pi fL,$$

where the unit of L is in henrys (H). Ohm's law relation is also applied to the inductor. In it, R is replaced by X_L such that

$$V = IX_L$$

Reactance is the opposition to the flow of a.c., offered by a capacitor or an inductor or both

Worked Example 5.3

Find the inductive reactance across an inductor of 0.2H inductance when an a.c. voltage of 60Hz is applied across it. If the voltage is given by $V = 150 \sin 120\pi t$, calculate the r.m.s. and peak values of the current.

Worked Example 5.4

The current amplitude in a pure inductor in a radio receiver is to be $250\mu A$ when the voltage amplitude is 3.60 V at a frequency of 1.60 MHz (at the upper end of the AM broadcast band). What inductive reactance is needed? What inductance?

5.5 Series containing R, L and C

Consider a series containing a resistance, an inductance and capacitance in series as shown in the figure below

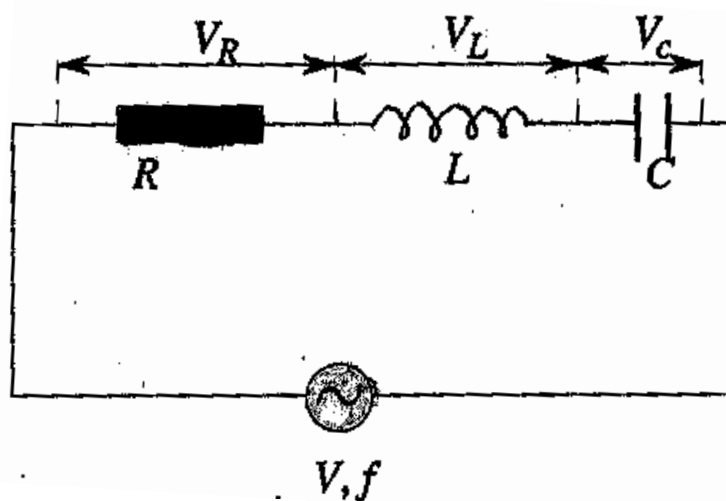


Figure 5.4: R, L, C circuit

The maximum peak of the current is given by

$$I_0 = \frac{V_0}{[R^2 + X^2]^{1/2}}, \quad X = X_L - X_C$$

Thus,

$$I_0 = \frac{V_0}{Z}, \quad \text{with } Z = [R^2 + X^2]^{1/2},$$

where Z is known as the *impedance* of the circuit

impedance (Z) is the overall opposition of a mixed circuit containing a resistor, an inductor and or a capacitor. It is measured in ohms.

Worked Example 5.5

A series circuit consists of a resistance 600ohms and an inductance of 5H. An a.c. voltage of 15volts (r.m.s.) and frequency 50Hz is applied across the series circuit. Calculate

- i. The current, I, flowing through the circuit
- ii. The voltage across the inductor
- iii. The phase angle between I and the applied voltage

- iv. The average power supplied
- v. The p.d. across the resistance.

5.6 Resonance in RLC series

Resonance is said to occur in an a.c. circuit when maximum current is obtained from such a circuit. That is, the maximum current is obtained when the impedance is minimum. This happens when $X_L = X_C$

The frequency at which resonance occurs is called resonant frequency (f_0). This is also the frequency at which $X_L = X_C$ or $2\pi f_0 L = \frac{1}{2\pi f_0 C}$. Thus, we can find

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

The resonance circuit finds applications in electronics. It is used to tune radios and TVs.

Worked Example 5.6

An a.c. voltage of amplitude 2.0 volts is connected to an RLC series circuit. If the resistance in the circuit is 5ohms, and the inductance and capacitance are 3mH and $0.05\mu F$ respectively, calculate

- i. The resonance frequency f_0
- ii. The maximum a.c. current at resonance.

CHAPTER SIX

6.1 Magnetic field

Magnetic field is defined as a region around a magnet in which the influence of the magnet can be felt or detected. Although this field cannot be seen but the effect can be felt demonstrated and mapped out using a compass needle. Such a needle when placed near the magnet will swing around and settle in a definite direction showing that a sort of force is present. Magnetic field is therefore an example of a *force field* and the force is called magnetic force.

Magnetic field is a vector field, i.e. such a field has both magnitude and direction. The compass needle is a sensitive instrument for finding the direction of the magnetic field. The direction of the field at any point is taken as the direction in which a N-pole placed at that point will tend to move.

Homework 2: describe the magnetic field around a straight conductor carrying current

Such a direction can always be obtained by applying the Fleming's right-hand grip rule or Clenched Fist rule:

If the straight wire is grasped with the right hand so that the thumb point in the direction of current then the direction in which the fingers are curled, indicate the direction of the magnetic field

Another alternative way of finding the field line direction is to use the *Maxwell's Corkscrew Rule*:

If a corkscrew is turned so that its tip travels along the direction of current, the direction of rotation of the corkscrew gives the direction of the magnetic field or magnetic lines of force.

6.2. Forces on a conductor

The strength of a magnetic field is usually measured in terms of a quantity called the magnetic flux density of the field, B . A conductor carrying an electric current, when placed in a magnetic field experiences a mechanical force. We can compute the force starting with magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on a single moving charge, where q is the charge and velocity with which the charge move is v .

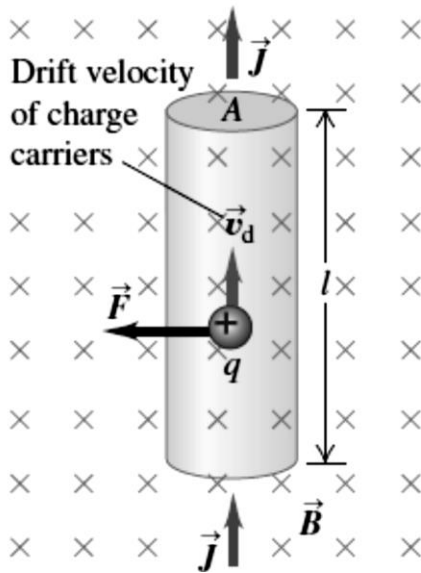


Figure 6.1: segment of a conducting wire

The above figure 6.1 shows the segment of a conducting wire, with length l and cross-sectional area A ; the current is from bottom to top. The average force on each charge is $\vec{F} = q\vec{v}_d \times \vec{B}$ directed to the left as shown in the figure; since $\vec{v}_d$ and $\vec{B}$ are perpendicular, the magnitude of the force is $F = qv_d B$.

The number of charges per unit volume is n ; a segment of conductor with length l has volume and contains a number of charges equal to nAl . The total force $\vec{F}$ on all the moving charges in this segment has magnitude

$$F = (nAl)(qv_d B) = (nqv_d A)(Bl)$$

Homework 3: Briefly explain drift velocity and current density. Hence, obtain an expression for current density in terms of drift velocity

The current density is $J = nqv_d$. The product JA is the total current I so we can rewrite above equation as

$$F = IBl$$

Suppose if the field $\vec{B}$ is not perpendicular to the wire but makes an angle ϕ with it, then, the magnetic force on the wire segment is then

$$F = IBl \sin \phi$$

Note: The greatest force occurs when $\theta = 90^\circ$, that is, when the conductor is at right angles to the field

The direction of force on a current carrying conductor placed perpendicular to the magnetic field is given by Fleming's left-hand which is stated as follows:

If the thumb, forefinger and middle finger are held mutually at right angle to one another with the **F**ore –finger pointing in direction of magnetic field, and the se**C**ond finger in direction of current, then the thu**M**b will point in direction of motion or force producing motion.

Worked example 6.1

A straight horizontal copper rod carries a current of 50.0 A from west to east in a region between the poles of a large electromagnet. In this region there is a horizontal magnetic field toward the northeast (that is, 45° north of east) with magnitude 1.20 T. (a) Find the magnitude and direction of the force on a 1.00-m section of rod. (b) While keeping the rod horizontal, how should it be oriented to maximize the magnitude of the force? What is the force magnitude in this case?

Worked example 6.2

A particle with a charge $-1.24 \times 10^{-8} \text{ C}$ of is moving with instantaneous velocity $\vec{v} = (4.19 \times 10^4 \text{ m/s})\hat{i} + (-3.85 \times 10^4 \text{ m/s})\hat{j}$. What is the force exerted on this particle by a magnetic field (a) $\vec{B} = (1.04 \text{ T})\hat{i}$ and (b) $\vec{B} = (1.04 \text{ T})\hat{k}$

6.3. Hall Effect

The **Hall effect** is the production of a voltage difference (the **Hall voltage**) across an electrical conductor, transverse to an electric current in the conductor and a magnetic field perpendicular to the current. It was discovered by Edwin Hall in 1879.

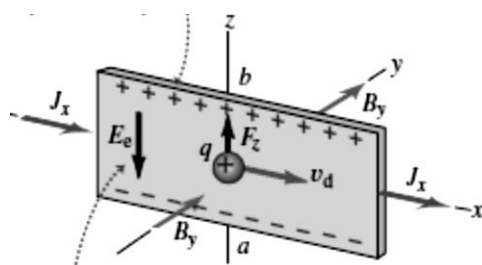


Figure 6.2

Homework 4: Describe Hall Effect by considering a conductor in the form of a flat strip.

This effect is described by considering a conductor in the form of a flat strip, as shown in figure 6.2 above. The magnetic field is in the +y-direction and we write it as B_y . The magnetic force (in the +z-direction) is $qv_d B_y$. The current density J_x is in the +x-direction. In the steady state, when the forces qE_z and $qv_d B_y$ are equal in magnitude and opposite

$$qv_d B_y + qE_z = 0 \quad \text{or} \quad E_z = -v_d B_y$$

This confirms that when q is positive, E_z is negative. The current density J_x is

$$J_x = nqv_d \text{ generally } J_x = n|q|v_d$$

Eliminating v_d between these equations, we find

$$nq = -\frac{J_x B_y}{E_z} \quad (\text{Hall Effect})$$

It should be noted that this result (as well as the entire derivation) is valid for both positive and negative q . When q is negative, E_z is positive, and conversely.

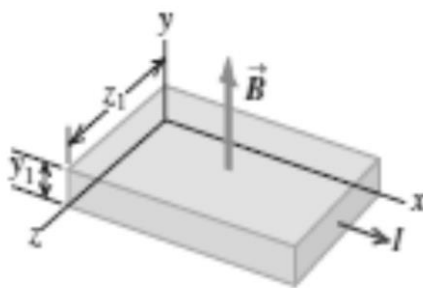
We can measure J_x , B_y and E_z so we can compute the product nq in both metals and semiconductors, q is equal in magnitude to the electron charge, so the Hall effect permits a direct measurement of n , the concentration of current-carrying charges in the material. The *sign* of the charges is determined by the polarity of the Hall emf, as we have described. The Hall Effect can also be used for a direct measurement of electron drift speed v_d in metals.

Worked example 6.3

You place a strip of copper, 2.0 mm thick and 1.50 cm wide, in a uniform 0.40-T magnetic field. When you run a 75-A current in the $+x$ –direction, you find that the potential at the bottom of the slab is $0.8\mu\text{V}$ higher than at the top. From this measurement, determine the concentration of mobile electrons in copper.

Worked example 6.4

Figure below shows a portion of a silver ribbon with $z_1 = 11.8\text{mm}$ and $y_1 = 0.23\text{mm}$ carrying a current of 120 A in the x -direction. The ribbon lies in a uniform magnetic field, in the y -direction, with magnitude 0.95T.



Apply the simplified model of the Hall effect. If there are 5.85×10^{28} free electrons per cubic meter, find (a) the magnitude of the drift velocity of the electrons in the x -direction; (b) the magnitude and direction of the electric field in the z -direction due to the Hall effect; (c) the Hall emf.

CHAPTER SEVEN

7.1. Electromagnetic induction

We now know that an electric current creates a magnetic field. The reverse effect of producing electricity by magnetism was discovered independently in 1831 by Faraday in England and Henry in America and is called electromagnetic induction. Indeed, the development of electrical engineering as we know it today began with Faraday and Henry who independently discovered the principles of induced e.m.f.s and the methods by which mechanical energy can be converted directly to electrical energy.

Electromagnetic induction is the production of electric current or voltage in a conductor whenever there is a relative motion between conductor and a field.

7.2. Laws of electromagnetic induction

There are two laws of electromagnetic induction: (1) Faraday's law and (2) Lenz's law.

(1) Faraday's law states that whenever there is a change in magnetic lines of force (e.m.f.) is induced, the strength of which is proportional to the rate of change of the flux linked with the circuit.

Or

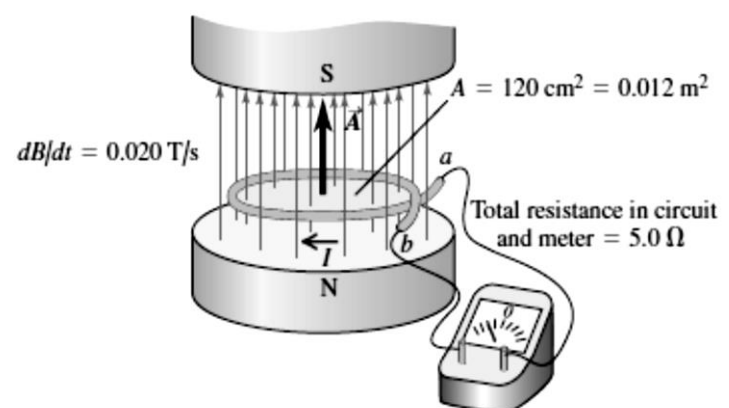
The induced e.m.f. in a closed loop equals the negative of the time rate of change of magnetic flux through the loop.

The magnetic flux or field lines linking a coil depend on (i) the magnetic field strength, (ii) the number of turns of the coil (iii) the area of each turn.

$$E = \frac{-d(N\Phi)}{dt} = \frac{-d(NBA \cos \phi)}{dt}$$

Worked example 7.1

The magnetic field between the poles of the electromagnet in Fig. shown is uniform at any time, but its magnitude is increasing at the rate of 0.02T/s The area of the conducting loop in the field is 120cm^2 , and the total circuit resistance,



including the $0.020 \text{ T}\cdot\text{s}$. meter, is 5.0Ω . Find the induced e.m.f. and the induced current in the circuit.

Worked example 7.2

A 500-loop circular wire coil with radius 4.00 cm is placed between the poles of a large electromagnet. The magnetic field is uniform and makes an angle of 60° with the plane of the coil; it decreases at 0.200 T/s . What are the magnitude and direction of the induced e.m.f. ?

(2) Lenz's law states that the direction of any magnetic induction effect is such as to oppose the cause of the effect.

Or

The induced e.m.f. is in such a direction to oppose or change the motion producing it.

7.3. E.m.f. induced in a straight conductor.

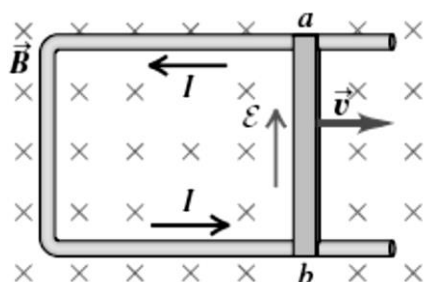
When a straight conductor is moved through a magnetic field an e.m.f. is induced between its ends. This movement must be in such a direction that the conductor cuts through the lines of magnetic flux, and will be a maximum when it moves at right angles to the field.

Let the length of the conductor be l and the flux density of the field be B . If the conductor moves with velocity v at right angles to the field then the flux cut per second will be Bvl (since the conductor will sweep out an area vl every second).

But the rate of cutting flux is equal to the e.m.f. induced in the conductor. Therefore If the conductor cuts through the flux at an angle θ the equation becomes

$$E = Blv \sin \theta$$

Worked example 7.3



The motional emf $\mathcal{E}$ in the moving rod creates an electric field in the stationary conductor.

Suppose the moving rod in Fig. shown is 0.10 m long, the velocity v is 2.5 m/s , the total resistance of the loop is 0.03Ω and B is 0.60 T . Find the motional e.m.f., the induced current, and the force acting on the rod.

7.4 Dynamo and Generators

A machine that converts mechanical energy into electrical energy or electrical energy into mechanical energy is called a *Dynamo*. When it changes mechanical energy into electrical energy, a dynamo is called a *Generator*; but when it changes electrical energy into mechanical energy, it is called a *Motor*.

7.4.1. Generators

There are two classes of generators; the Alternating Current (A.C.) generator, and the Direct Current (D.C.) generator.

The a.c. generator or alternator

A simple form of the a.c. generator is shown in the figure 7.1 (a) below. A coil (the rotor) is rotated between the poles of a d.c. electromagnet (energised by the **field coils**), except in the case of a bicycle dynamo where a permanent magnet is used, and the e.m.f. generated is taken from the ends of the coil. These are connected to sliding contacts known as **slip rings** on the axle, and contact is made with these by two pieces of carbon (the **brushes**) which press against the slip rings. As the coil rotates it cuts through the lines of magnetic flux producing an induced e.m.f. A much smoother output is obtained by having a number of coils wound on an iron core which is laminated to reduce eddy currents. The output of such a generator is shown in figure

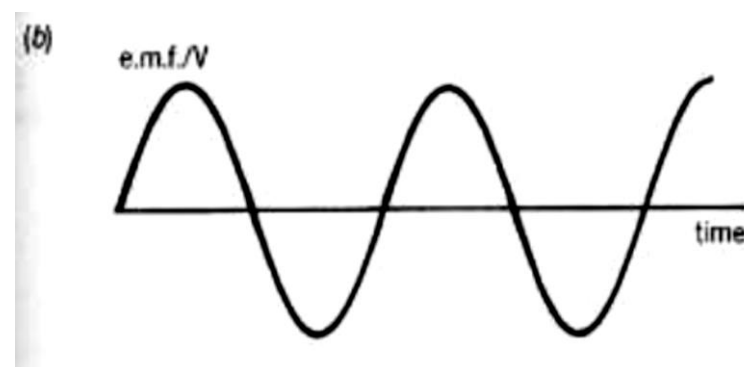
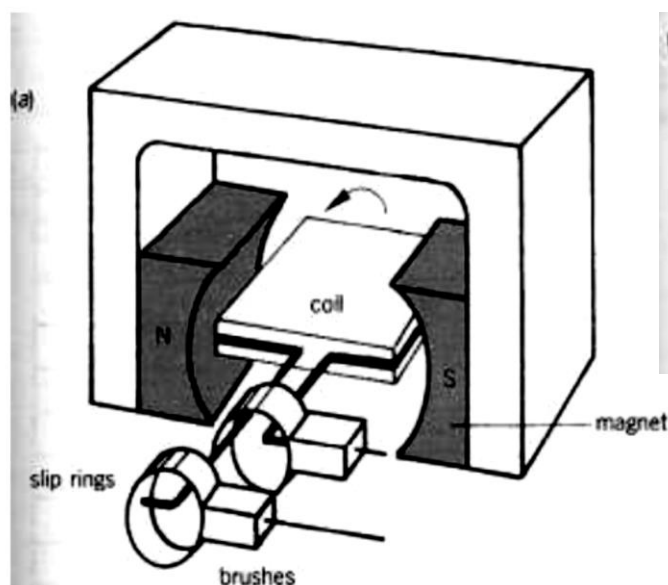


Figure 7.1

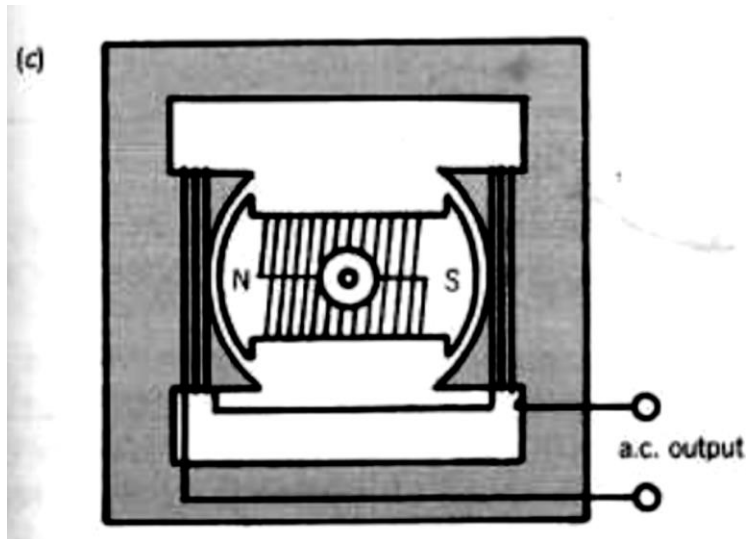
in figure 7.1 (c). The advantage of this is that the slip rings and brushes have to carry only the small current needed to magnetize the rotating electromagnet while the current produced the static field coils may be many hundreds of amps. In fact in

7.1 (b). In generators where the output current may be very large, as in a power station, it is the magnet that

rotates while the coil remains at rest.

A simplified version of this is shown

modern alternators installed in a power station the e.m.f. generated will be some 25 kV and the current produced over 1000 A!



The e.m.f. generated in a coil is given as $E = BAN\omega$ where B is flux density, A is area, N is number of turns and ω is the angular velocity.

The d.c. generator

In the d.c. generator the output from the rotor assembly is fed to a commutator where the brushes press against a split ring of copper. This means that a varying but unidirectional e.m.f. will be produced. A d.c. generator and its output is shown in simplified form in Figure below. As with the a.c. generator, the d.c. machine usually uses rotating field coils, a series of them being

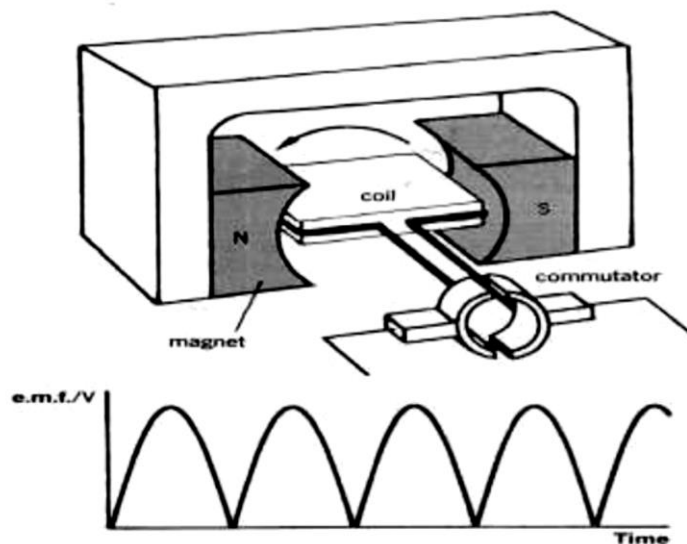


Figure 7.2

wound in slots in the core; the rotating coils and the core are known as the armature. The output is then much steadier, a ripple effect being obtained. The d.c. generator may be made 'self-exciting' by putting the field coils and armature in series or parallel, the current required for the field coils being produced by the generator itself. There is nearly always some residual magnetism in the core of the armature to aid the starting of such a generator.

7.5. The transformer

Transformer is an electrical device for changing the size of an a.c. voltage. It acts to increase or decrease the e.m.f. of an alternating current. The transformer uses the property of mutual inductance to change the voltage of an alternating supply. It may be used in the home to give a low-voltage output from the mains for a cassette

recorder or train set, or in a power station to produce very high voltages for the National Grid.

In its simplest form it consists of two coils known as the primary and secondary, wound on a laminated iron former that links both coils (See Figure below). The core must be laminated otherwise large eddy currents would flow in it. The laminations are usually E-shaped, and the primary and secondary are wound one on top of the other to improve magnetic linkage.

Homework 5: explain what is meant by eddy current.

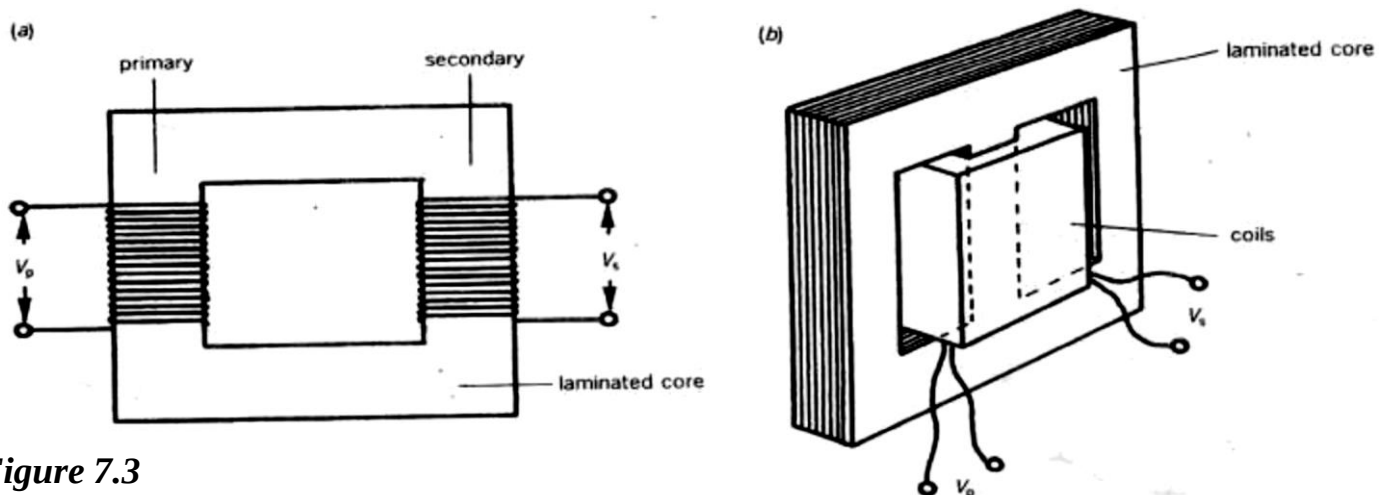


Figure 7.3

An a.c. voltage is applied to the primary coil and this produces a changing magnetic field within it. This changing magnetic field links the secondary coil and therefore induces an e.m.f. in it. The magnitude of this induced e.m.f. (V_s) is related to the e.m.f. applied to the primary (V_p) by the equation:

$$\frac{V_s}{V_p} = -\frac{n_s}{n_p}$$

where n_p and n_s are the number of turns on the primary and secondary coils respectively. The negative sign means that the voltage induced in the secondary is 180° (or π) out of phase with that in the primary. If the output voltage is greater than the input voltage the transformer is known as a **step up** transformer and if the reverse is true it is called a **step down** transformer.

The current in the secondary coil produces its own magnetic flux, which is opposite to that of the primary. When the current in the secondary is increased by increasing the load the flux in the core is reduced. The back e.m.f. in the primary therefore falls and the current in the primary increases. Eventually the situation will stabilize.

The output voltage may be measured with a meter but a better method is to use an oscilloscope since it draws no current from the transformer.

We have assumed here that there is no leakage of flux that is, that all the flux produced by the primary links the secondary and that there are no energy losses. In practice, however, energy is lost from a transformer in the following ways:

- (a) heating in the coils - this can be reduced by keeping their resistance low;
- (b) eddy current losses in the core; reduced by the laminated core already mentioned;
- (c) hysteresis loss - every time the direction of the magnetizing field is changed some energy is lost due to heating as the magnetic domains in the core realign. This is reduced by using a 'soft' magnetic material for the core such as permalloy or silicon-iron. For soft magnetic materials the loss might be about 0.02 J per cycle.

Despite these energy losses transformers are remarkably efficient (up to 98 per cent efficiency is common and they are in fact among the most efficient devices ever developed.

If we now assume the transfer of energy from primary to secondary to be 100 per cent efficient, then power in primary = power in secondary and therefore:

$$\frac{I_s}{I_p} = -\frac{n_s}{n_p}$$

and so a step up transformer for voltage will be a step down transformer for current, and vice versa. Thus, the efficiency of a transformer is defined by

$$\text{Efficiency} = \frac{\text{Output power}}{\text{Input power}} \times 100\% = \frac{\text{Power in secondary coils}}{\text{Power in primary coils}} \times 100\%$$

Worked example 7.1

A transformer supplies 15V from 220V mains. If the transformer takes 0.7A from the mains when used to light three lamps connected in parallel and each rated 15V, 40W, calculate (a) the efficiency of the transformer (b) the cost of using it for 24hrs at 30k per kWh.

Worked example 7.2

(a) Draw a label diagram to explain the working of a transformer which can produce 24V from a 240V supply. (b) if the efficiency of the transformer is 80% and the current in the secondary is 10A, calculate the current in the 240V supply.

7.6. Magnetic properties of materials

All matter exhibits magnetic properties when placed in an external magnetic field. Even substances like copper and aluminum that are not normally thought of as having magnetic properties are affected by the presence of a magnetic field such as that produced by either pole of a bar magnet. Depending on whether there is an attraction or repulsion by the pole of a magnet, matter is classified as being either **paramagnetic** or **diamagnetic**, respectively. A few materials, notably iron, show a very large attraction toward the pole of a permanent bar magnet; materials of this kind are called **ferromagnetic**.

In 1845 Faraday became the first to classify substances as either diamagnetic or paramagnetic. He based this classification on his observation of the force exerted on substances in an inhomogeneous magnetic field. At moderate field strengths, the magnetization $\mathbf{M}$ of a substance is linearly proportional to the strength of the applied field $\mathbf{H}$. The magnetization is specified by the magnetic susceptibility χ , defined by the relation $\mathbf{M} = \chi\mathbf{H}$. A sample of volume V placed in a field $\mathbf{H}$ directed in the x -direction and increasing in that direction at a rate $\frac{dH}{dx}$ will experience a force in the x -direction of $\mathbf{F} = \chi\mu_0 V H \left(\frac{dH}{dx} \right)$. If the magnetic susceptibility χ is positive, the force is in the direction of increasing field strength, whereas if χ is negative, it is in the direction of decreasing field strength. Measurement of the force $\mathbf{F}$ in a known field $\mathbf{H}$ with a known gradient $\frac{dH}{dx}$ is the basis of a number of accurate methods of determining χ .

Substances for which the magnetic susceptibility is negative (e.g., copper and silver) are classified as **diamagnetic**. The susceptibility is small, on the order of -10^{-5} for solids and liquids and -10^{-8} for gases. A characteristic feature of diamagnetism is that the magnetic moment per unit mass in a given field is virtually constant for a given substance over a very wide range of temperatures. It changes little between solid, liquid, and gas; the variation in the susceptibility between solid or liquid and gas is almost entirely due to the change in the number of molecules per unit volume. This indicates that the magnetic moment induced in each molecule by a given field is primarily a property characteristic of the molecule.

Substances for which the magnetic susceptibility is positive are classed as **paramagnetic**. In a few cases (including most metals), the susceptibility is independent of temperature, but in most compounds it is strongly temperature dependent, increasing as the temperature is lowered. Measurements by the French physicist Pierre Curie in 1895 showed that for many substances the susceptibility is inversely proportional to the absolute temperature T ; that is, $\chi = C/T$. This approximate relationship is known as Curie's law and the constant C as the Curie constant. A more accurate equation is obtained in many cases by modifying the above equation to

$= \frac{C}{T - \theta}$, where θ is a constant. This equation is called the Curie–Weiss law (after Curie and Pierre-Ernest Weiss, another French physicist). From the form of this last equation, it is clear that at the temperature $T = \theta$, the value of the susceptibility becomes infinite. Below this temperature, the material exhibits spontaneous magnetization *i.e.*, it becomes ferromagnetic. Its magnetic properties are then very different from those in the paramagnetic or high-temperature phase. In particular, although its magnetic moment can be changed by the application of a magnetic field, the value of the moment attained in a given field is not always the same; it depends on the previous magnetic, thermal, and mechanical treatment of the sample

CHAPTER EIGHT

Introduction to Modern Physics

The term **modern physics** refers to the post-Newtonian conception of physics. Modern physics deals with the underlying structure of the smallest particles in nature ("quantum" mechanics), as well as a rigorous understanding of the fundamental interaction of particles, understood as forces. Modern physics often involves extreme conditions; quantum effects usually involve distances comparable to atoms (roughly 10^{-9} m), while relativistic effects usually involve velocities comparable to the speed of light (roughly 10^8 m/s).

The term "modern physics" implies that classical descriptions of phenomena are lacking, and that an accurate, "modern", description of reality requires theories to incorporate elements of quantum mechanics or Einstein relativity, or both. For example, when analyzing the behavior of a gas at room temperature, most phenomena will involve the (classical) Maxwell–Boltzmann distribution. However near absolute zero, the Maxwell–Boltzmann distribution fails to account for the observed behavior of the gas, and the (modern) Fermi–Dirac or Bose–Einstein distributions have to be used instead.

In general, the term is used to refer to any branch of physics either developed in the early 20th century and onwards, or branches greatly influenced by early 20th century physics.

Very often, it is possible to find – or "retrieve" – the classical behaviour from the modern description by analyzing the modern description at low speeds and large distances (by taking a limit, or by making an approximation). When doing so, the result is called the *classical limit*.

These are generally considered to be the topics regarded as the "core" of the foundation of modern physics:

- Atomic theory and the evolution of the atomic model in general
- Radioactivity: fusion, fission and half-life
- Photoelectric effect
- Wave–particle duality and uncertainty principle
- Black body radiation

- Franck–Hertz experiment
- Geiger–Marsden experiment (Rutherford's experiment)
- Gravitational lensing
- Michelson–Morley experiment
- Quantum thermodynamics
- Perihelion precession of Mercury
- Stern–Gerlach experiment
- Wave–particle duality and uncertainty principle.

These topics shall be treated in PHY 212 and PHY 313. However, the brief overview of the first four topics shall be treated here.

8.1. Atomic Theory

In chemistry and physics, **atomic theory** is a scientific theory of the nature of matter, which states that matter is composed of discrete units called atoms. In the days of Sir Isaac Newton, the atom was known to be a tiny, hard, indestructible sphere. Although this model provided a good basis for the kinetic theory of gases, new models had to be devised when experiments revealed the electrical nature of atoms.

8.1.1. Thompson's Model

J.J. Thomson suggested a model that describes the atom as a volume of positive charge with electrons embedded throughout the volume, much like the seeds in a watermelon or raisins in thick pudding

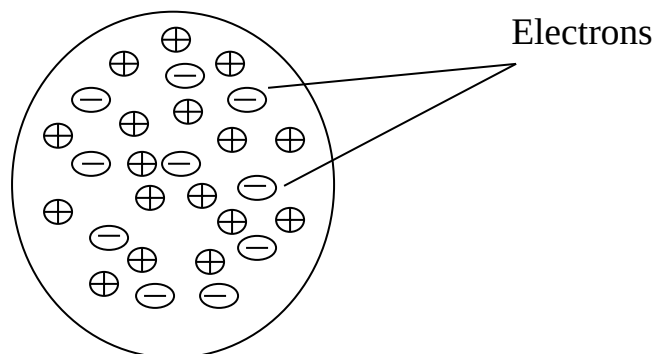


Figure 8.1: Diagram showing Thomson's model of the atom

8.1.2. Rutherford's Model of the Atom

A careful experimental study of the scattering of α – *particles* by Geiger and Marsden provided information about the distribution of mass and charge within the atom and in 1911 led to Rutherford's model of the atom. In this, he supposed that most of the mass of the atom is concentrated in the nucleus at its centre which also carries the positive charge, the diameter being about $10fm$ ($10^{-14}m$). This is surrounded by an electron cloud extending out beyond $0.1nm$ and which makes the whole atom electrically neutral.

In the experiment by Geiger and Marsden, α – *particles* from a source S (figure 8.2) were restricted to a narrow pencil by a hole H in the screen and fell upon a piece of metal foil F. the scattered α – *particles* were detected by a zinc sulphide screen Z, the individual scintillations being observed and counted using the microscope M. Foils of aluminum, silver and gold were used. In all these experiments;

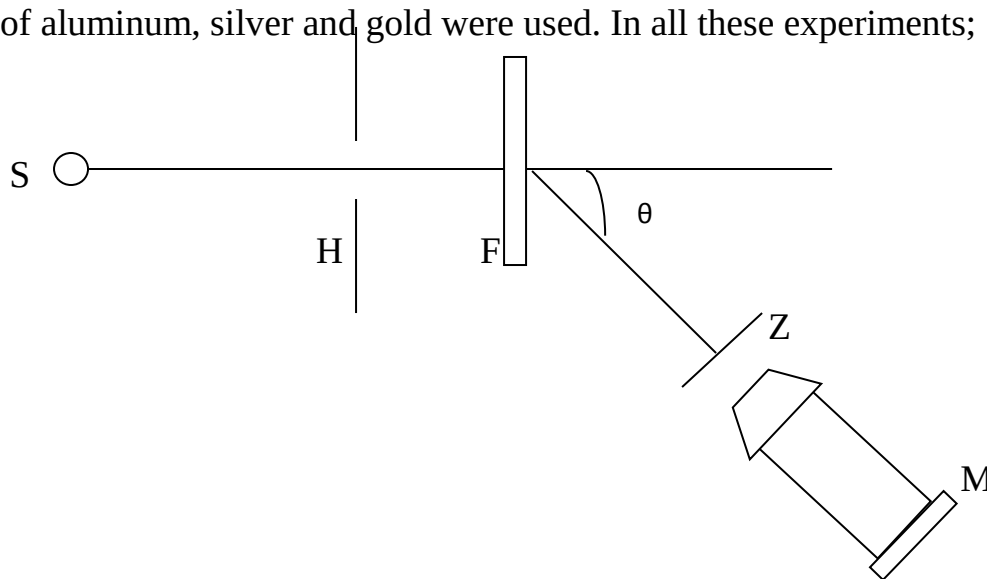


Figure 8.2: Scattering of α – *particles* by thin metal films

- i. Almost all the α – *particles* passed straight through the metal foil and were deflected only slightly. This showed that the atom was largely empty space.
- ii. A small fraction was so strongly deviated that the α – *particles* emerged again on the same side of the foil as they entered, showing that some of the α – *particles* were deflected through angles greater than 90° , i.e reflected back towards the source. This suggested that somewhere within the atom, there was a very small massive particle carrying a positive charge so that the

charge on the atom as a whole was zero. This small massive particle was called the nucleus.

Rutherford's model is also called the planetary model; here the atom is viewed in analogy with the solar system. In few words, in Rutherford's model, the atom is said to have a positively charged, centrally located nucleus with electrons revolving in circular orbits.

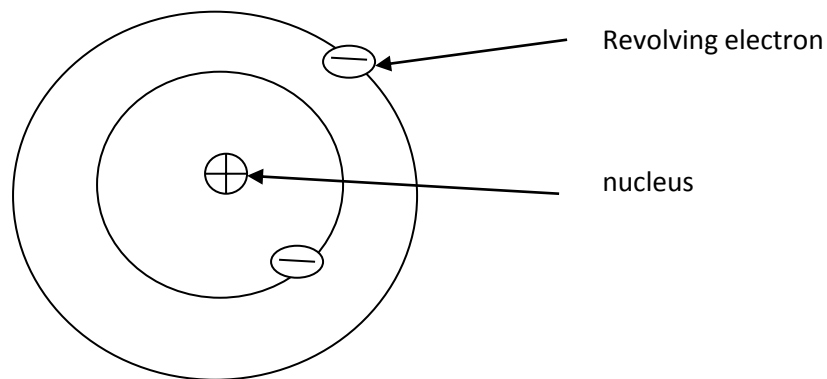


Figure 8.3: *Rutherford's planetary Model*

8.2. Radioactivity

Radioactivity is the spontaneous disintegration of unstable atomic nuclei with the emission of particles and electromagnetic radiation. This disintegration of the nucleus is called nuclear decay.

The study of natural radioactivity began accidentally in 1896, one year after Rontgen discovered x-rays. Henri Becquerel discovered radiation from uranium salts that seemed similar to x-rays. Intensive investigation decades after by the Curies, Rutherford and others revealed that the emissions consist of positively and negatively charged particles and neutral rays; they were given the names alpha, beta and gamma particles respectively, because of their differing characteristics.

The decaying nucleus is usually called the parent nucleus; the resulting nucleus (as a result of the decay) is the daughter nucleus. When a radioactive nucleus decays, the daughter nucleus may also be unstable. In such a case, a series of successive decays occurs until a stable configuration is reached. Several such series are found in nature. The most abundant radioactive nuclide found on earth is the Uranium isotope ^{238}U ,

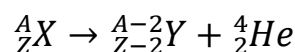
which undergoes a series of 14 decays including 8 alpha emissions and six beta emissions before terminating at a stable isotope of lead ^{206}Pb .

8.2.1. Types of radioactive decay

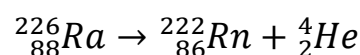
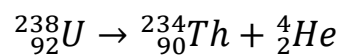
Nearly 90% of the known 2500 nuclides are radioactive in nature. When unstable nuclides decay they usually emit particles and radiation. The different decay processes are described:

1. Alpha decay

An alpha (α) decay occurs principally with nuclei that are too large to be stable. An alpha particle is a ^4He nucleus. If a radioactive nucleus emits an alpha particle, it loses two neutrons (N) and two protons (Z) - the decay process for alpha particles is:



where X is the parent nucleus, Y is the daughter nucleus and $A=N+Z$. Familiar examples of alpha emitters are Uranium decaying to Thorium and Radium decaying to Radon;

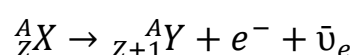


When one element changes into another, as happens in the alpha decay, the process is called spontaneous decay or transmutation. Alpha particles are deflected slightly in an electric or magnetic field. They are always emitted with definite kinetic energies determined by the conservation of momentum and energy. Their penetrating power is very low and they can travel only several centimeters in air, being stoppable by a thin sheet of aluminium or paper.

2. Beta decay

There are three processes of the beta (β) decay:

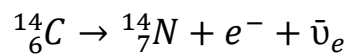
- i. Beta negative decay: a radioactive nucleus undergoes β^- decay when its atomic number increases by one through the addition of one proton, and at the same time, one neutron is lost so the mass of the daughter isotope is the same as the parent isotope;



In the β^- decay, we expect the electron to carry away most of the kinetic energy as opposed to the daughter nucleus, but experiments showed that some of the energy was missing. It turns out that a third particle is being released during the beta decay; it is called the neutrino, as named by Enrico Fermi and proposed by Wolfgang Pauli - the neutrino is given the symbol ν , and the antineutrino is given the symbol $\bar{\nu}_e$. The neutrino has the following properties:

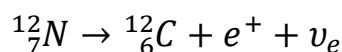
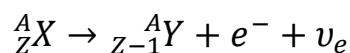
- zero electric charge
- a mass much smaller than the electron, but not zero
- very weak interaction with matter (very hard to detect)

In the β^- decay, an electron and an antineutrino are emitted. A good example of this decay process is seen in the radioactive decay of carbon-14;



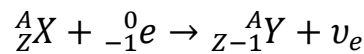
Beta negative decay usually occurs with nuclides for which the neutron-to-proton ratio, N/Z is too large for stability. Beta negative particles are electrons capable of travelling at speeds approaching the speed of light. Their low mass allows them to be deflected greatly in an electric or magnetic field in the opposite direction as the deflection of α -particles. Their high speeds give them greater penetrating power than the α -particles. Using the conservation of mass-energy, the β^- decay can occur whenever the neutral atomic mass of the parent atom is larger than that of the daughter atom.

- Beta positive decay: in this reaction a positron is emitted; a positron is exactly like an electron in mass and charge force except with a positive charge. A positron is formed when a proton breaks into a neutron with mass and no charge and a positron with no mass and the positive charge. A positron emission is common in lighter elements where the N/Z ratio is too small for stability. Thus in a β^+ decay, a positron and a neutrino are emitted in this process;



β^+ decay can occur whenever the atomic mass of the parent atom is at least two electron masses larger than that of the daughter atom.

- iii. Electron Capture: in this decay process a nucleus captures one of its own atom's inner shell electrons which reduce the atomic number by one. This captured electron combines with a proton in the nucleus to form a neutron and a neutrino is emitted. This electron capture process is common in larger elements with a low N/Z ratio.



The electron capture process can occur whenever the neutral atomic mass of the parent atom is larger than that of the parent atom.

3. Gamma decay

The process of gamma (γ) decay occurs when a nucleus is left in an excited state, either by bombardment with high energy particles or a radioactive transformation, and decays to the ground state with the emission of one or more photons with typical energies of 10KeV to 5MeV. These photons are called gamma rays or gamma photon rays. This gamma decay does not result in any changes in the composition of the atomic nuclide. They are high energy electromagnetic radiation with short wavelengths. They have the highest penetrating power, and are able to penetrate at least 30cm of lead.

8.2.2 Radioactive Half-Life and Activity

The half-life $T_{\frac{1}{2}}$ of a radioactive nucleus is the time required for the nuclei to decay or disintegrate to half its original number. The half-life is independent of the physical state, temperature, and chemical compound the nucleus finds itself in. The prediction of the decay can be stated in terms of the half-life and decay constant;

$$T_{\frac{1}{2}} = \frac{0.693}{\lambda} \quad (32)$$

where λ is the decay constant.

The mean lifetime, T_{mean} generally called the lifetime of a nucleus or unstable particle is proportional to the half-life $T_{\frac{1}{2}}$;

$$T_{mean} = \frac{T_{\frac{1}{2}}}{0.693} \quad (33)$$

A common unit of activity is the curie, abbreviated Ci, which is defined to be 3.70×10^{10} decays per second. This is approximately equal to the activity of one gram of radium. The SI unit of activity is the becquerel (Bq). One becquerel is one decay per second, so;

$$1\text{Ci} = 3.70 \times 10^{10}\text{Bq} = 3.70 \times 10^{10}\text{decays/s}$$

8.2.3. Law of radioactive decay

Since the decay of a radioactive nucleus is a purely statistical process; there is no way to predict when any individual nuclei will decay. The rate of decay varies over an extremely wide range for different nuclides.

Let $N(t)$ be the very large number of radioactive nuclei in a sample at time t , and let $dN(t)$ be the negative change in that number during a short time interval dt . The number of decays during the interval dt is $-dN(t)$. The rate of change of $N(t)$ is the negative quantity $\frac{dN(t)}{dt}$.

Thus $-\frac{dN(t)}{dt}$ is called the decay rate or activity of this sample specimen. The larger the number of nuclei in the specimen, the more nuclei decays during any time interval. That is, the activity is directly proportional to $N(t)$;

$$-\frac{dN(t)}{dt} \propto N(t)$$

$$\therefore, -\frac{dN(t)}{dt} = \lambda N(t)$$

where λ is a constant called the decay constant, and it has different values for different nuclides. A large value of λ corresponds to rapid decay, while a small value corresponds to a slower decay. λ can be interpreted as the probability per time that any individual nucleus will decay. Now, re-arranging, we have

$$\frac{dN(t)}{N(t)} = -\lambda dt$$

Integrating between the limits $N(t) = N_0$, and $N(t) = N$ for the number of nuclei at time $t=0$, gives;

$$\ln N_0 - \ln N = \lambda t$$

Taking the anti-logarithm of both sides of the equation above, we have;

$$N = N_0 e^{-\lambda t}$$

This result is the law of radioactive decay.

Since N is directly proportional to the activity (A) and the mass (m) of the nuclei sample, we can express the law of radioactive decay (equation 4) in three alternative forms;

- a. Number of nuclei remaining;

$$N = N_0 e^{-\lambda t}$$

- b. Activity;

$$A = A_0 e^{-\lambda t}$$

- c. Mass;

$$m = m_0 e^{-\lambda t}$$

Worked Example 8.1.

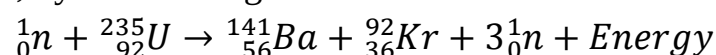
The radioactive isotope ^{57}Co decays by electron capture with a half-life of 272 days. (a) Find the decay constant and the lifetime. (b) if you have a radiation source containing ^{57}Co , with activity $2.00\mu\text{Ci}$, how many radioactive nuclei does it contain? (c) What will be the activity of your source after one year?

8.2.4. Nuclear Energy

The protons and neutrons (nucleons) in the nucleus of each atom are held together by very powerful nuclear forces. An enormous amount of energy is therefore required to tear the nucleon apart. This energy is over 10^6 times more than that required to remove the electrons from an atom.

1. Nuclear Fission

Nuclear fission is the splitting up of the nucleus of a heavy element into two approximate equal parts with the release of a huge amount of energy and neutrons. For example the heavy nucleus of Uranium-235 can split into two other elements, Krypton and barium, by bombarding it with a slow neutron



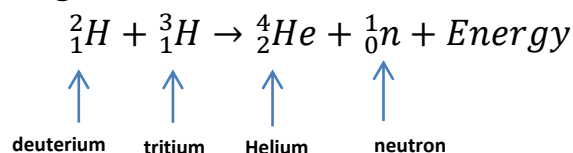
It was found that the total mass of the component products is less than the mass of the original Uranium. The difference is (*mass defect*) is a measure of the nuclear energy released. Accordingly to Albert Einstein

$$E = \Delta mc^2$$

where E is the energy release, Δm is the difference in mass and c is the speed of light.

2. Nuclear Fusion

Nuclear fusion is a nuclear process in which two or more light nuclei combine or fuse to form a heavier nucleus with release of a large amount of energy. A typical example is the fusion of two hydrogen nuclei to form Helium, He.



Homework 6: what are the advantages of fusion over fission

8.3. The Photo-electric effect

When light of sufficiently high frequency (particularly ultraviolet light) is shined on a metal surface, electrons are emitted from the metal surface. This phenomenon is called photoelectric effect. The emitted electrons are called photoelectrons.

8.3.1. Fact about Photo-electric effect

- (a) Emission of the electrons is practically instantaneous, no matter how weak the intensity of the incident light
- (b) There is a threshold frequency f_0 for the emission that depends on the material that is illuminated. If the frequency ν of the incident radiation is less than the critical value ν_0 , there will be no emission no matter how intense the illumination. However if $\nu > \nu_0$, there will be emission and the kinetic energy of the emitted electrons spans a finite range, up to a maximum value that is a linear function of the frequency of the incident light.
- (c) Provided that $\nu > \nu_0$ the number of electrons emitted per second is proportional to the intensity of the incident radiation.

According to Einstein, the maximum kinetic energy required for liberation of an electron is

$$\frac{1}{2}mv_{max}^2 = hf - W_0 = hf - hf_0$$

where W is the work function⁴, f_0 is the threshold frequency.⁵

Worked example 8.2

Compute the frequency of the photon whose energy is required to eject a surface electron with a kinetic energy $3.5 \times 10^{-16} \text{ eV}$ if the work function of the metal is $3.5 \times 10^{-16} \text{ eV}$. Take $h = 6.6 \times 10^{-34} \text{ Js}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

8.4. The Uncertainty Principle

The conceptual difficulties in accepting the wave-particle dualism arise because popular ideas of waves and particles are based upon large-scale observations of large-scale natural phenomena and experiments.

If measurement is to be taken for the position and speed of a particle at any instant, experimental uncertainties will affect the values of the observables. According to classical mechanics, no fundamental barrier to an ultimate refinement of the apparatus or experimental procedures exists. In other words, it is possible to make such measurements but with arbitrary small uncertainty.

In 1927, Werner Heisenberg introduced the notion of the existence of measurement and procedure barrier, which is now known as the uncertainty principle.

This principle states that, the momentum and position of a particle cannot simultaneously be measured with complete certainty. In other words, if a measurement of position of a particle is made with precision Δx , and a simultaneous measurement of linear momentum is made with precision Δp , then the product of the two uncertainties can never be smaller than $\frac{\hbar}{2}$.

i.e.,

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

where $\hbar = \frac{h}{2\pi}$

If Δx is very small, then Δp will be large, and vice versa.

⁴ The work function (W) of a metal is the minimum energy required to liberate an electron from a metal surface.

⁵ Threshold frequency is the frequency of light which, falling on a surface, is just sufficient to liberate electrons without giving them additional kinetic energy.

According to Heisenberg, the uncertainties Δx and Δp do not arise from imperfections in the measuring instruments. Rather, he pointed that they arise from the quantum structure of matter.

The Uncertainty principle also applies to the measurement of the energy E with precision ΔE and time t with precision Δt as follows

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

This relation is derivable from the momentum-position uncertainty relation. Similarly, it applies to the measurement of angular momentum L of a body and its angular position ϕ , as above;

$$\Delta L \Delta \phi \geq \frac{\hbar}{2}$$

Worked example 8.3

The speed of an electron is measured to be $3.0 \times 10^3 \text{ ms}^{-1}$ to an accuracy of 0.008%. Find the minimum uncertainty in determining the position of the electron.

ADVICE

- ***Take responsibility for yourself.*** When some students begin University life their sense of personal responsibility seems to disappear. Parents or guardians are no longer “cracking the whip” making certain that everything is getting done correctly or on time. That work is now the responsibility of the student. Failure to recognize this fact has resulted in even some of the best secondary school students failing as university students.
- ***Set high personal and academic standards for yourself, and live up to them.*** Listen to that little voice inside you that says, “I can do this.” Believe in yourself. Realize that school is work; it’s not play time. Settle for nothing less than your very best. Willingness to accept anything less than the very best too often becomes a self-fulfilling prophesy. Strive for an “A” in all your courses. If you fall short of an “A”, you might earn a “B”. If you fall short of a “C”, you might earn a “D” or “E”.
- ***Honesty is the best policy.*** Avoid cheating in all its forms – collusion, plagiarism, copying, etc. Students who cheat seriously fail to learn what is oftentimes important, and this doesn’t help them in the long run. Sometimes the only things they do learn – after getting caught – is that cheating doesn’t pay.
- ***Don’t put off until tomorrow what you can do today.*** Work should come before pleasure. Manage your time effectively; set up a timeline for getting work completed in each of your courses. Set aside adequate time for homework, study, sleep, relationships, and work. You need not always finish every task all at once. Remember, you can write a 365-page book every year if you only write one page per day.
- ***Strive to understand.*** Don’t merely memorize; increase your depth of understanding. You need to attempt to fully comprehend what you need to know and be able to do as a result of your education.

- ***Don't ignore or deny your personal and academic problems.*** Problems will often get worse if they are not directly addressed in a timely fashion. Procrastination in any of its many forms can lead to a small problem getting much worse. Get help when you need it. Speak to your course instructors, your advisor, or your parents.
- ***Make yourself a well-rounded person.*** Consider all four dimensions of life as you strive to educate yourself – physical, spiritual, intellectual and social. Spend time each day developing each of these four dimensions.

I wish you the best

Falaye, B.G.